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QLD CRASH INCIDENTS REPORT VISUALISATION

BUSA8090

DATA AND VISUALISATION FOR BUSINESS

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1. Data Exploration

The dataset “Crash data from Queensland roads,” provided by the Queensland Government, contains multi-year records of traffic crashes across the state. Each row represents a single crash event, capturing attributes related to location, severity, conditions, casualties, and vehicle involvement.

The dataset enables analysis of crash trends, contributing factors, and spatial-temporal patterns. Key attributes include:

- Crash date and time (for time-series analysis)
- Severity (fatal, hospitalisation, medical treatment, minor injury)
- Location data (suburb, region)
- Road/environmental conditions (lighting, weather, speed)
- Vehicle type counts (car, motorcycle, bus, etc.)

2. Abnormal Data Points

Table 1: Abnormal Data Points

Attribute Name	Example and Explanation	How to handle
Crash_Time	00:00 paired with Daylight, Dawn/Dusk lighting Midnight time appears to be a default placeholder for missing or unrecorded values. Lighting context further suggests time inaccuracy.	186 affected rows (<0.05%) were removed due to unreliability in time-based visualisation.
Crash_Longitude/ Crash_Latitude	Empty/null Missing geographic information.	598 missing rows were removed. Since the proportion is less than 1% of the dataset, removal ensured data consistency.
Crash_Speed_Limit	Empty/null Indicates missing speed limit information.	3 missing rows were removed. Since the proportion is less than 1% of the dataset, removal ensured data consistency.
Crash_Road_Surface_Condition/ Crash_Atmospheric_Condition/	Unknown Some entries listed as "Unknown".	We retained them for general analysis but filtered out in condition-specific visualisations.
Crash_Lighting_Condition	Unknown, or logically inconsistent with time (e.g. 00:00 + Daylight/Dawn)	Inconsistent records were removed. 'Unknown' retained for general analysis but filtered out in condition-specific visualisations.
Crash_Severity	Property Damage Only This severity type was no longer recorded after 2010, likely due to reporting policy changes.	We decided to exclude "Property Damage Only" from all crash severity analysis across the full period to maintain consistency and comparability

3. Data Transformation

To prepare the dataset for use in Tableau, I conducted the following transformations in Excel.

- **Crash Time Filtering:** Removed 186 records where crash time was 00:00 with daylight or dawn lighting, to ensure accuracy in time-based analysis.
- **Geographic Filtering:** Removed 598 records with missing value to ensure map accuracy.
- **Speed Limit Filtering:** Removed 3 records with missing value to ensure data accuracy.

4. Visualisation Analysis and Findings

4.1 Time series visualisation

Yearly Crash Trend in Queensland (2001–2023),
with Severity Breakdown

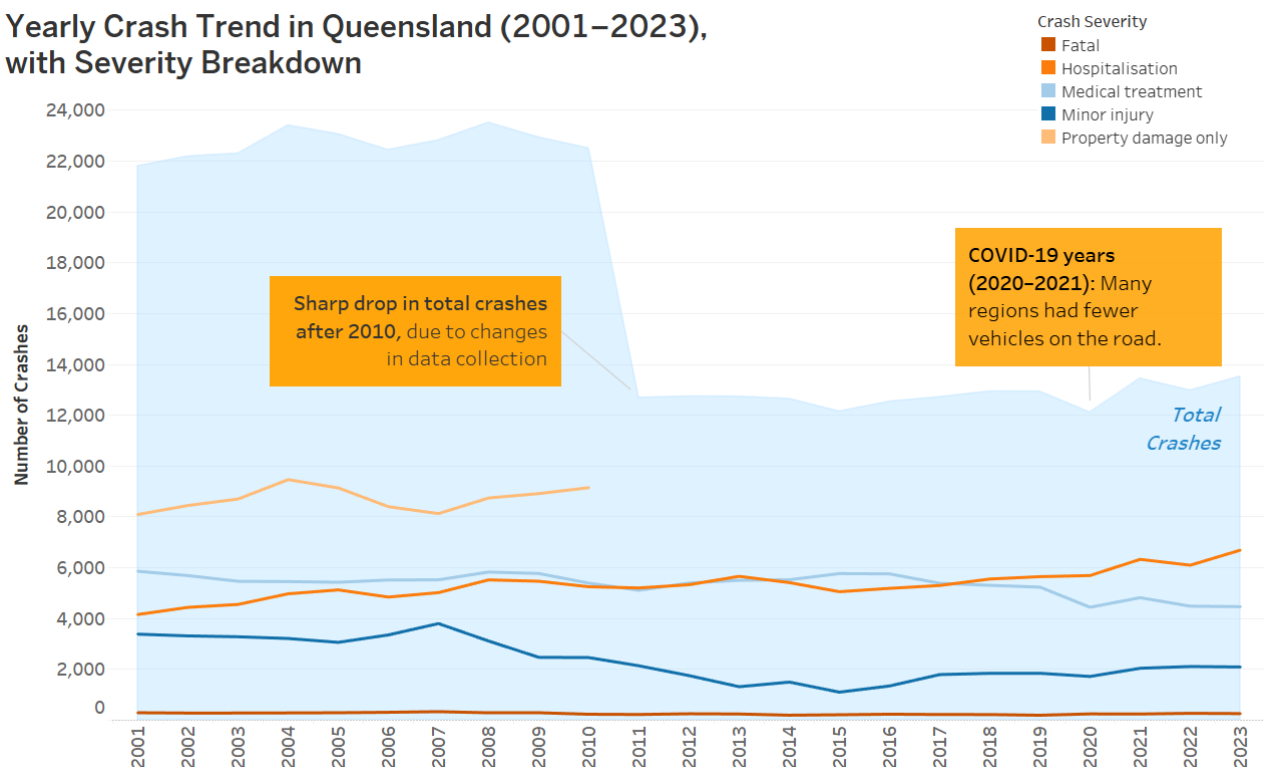


Figure 4.1. Road Crash Time Series by Severity in Queensland (2001-2023)

This time-series analysis explores road crash trends in Queensland from 2001 to 2023, revealing shifts in both volume and severity. From 2001 to 2010, annual crash counts remained stable at around 22,000-24,000. A sharp drop occurred post-2010, likely due to changes in data reporting rather than improved road safety. Another notable dip appears during the COVID-19 years (2020-2021), when traffic volumes declined.

In terms of severity, fatal crashes have remained consistently low, indicating that the most serious outcomes are rare. Property damage-only crashes were common before 2010 but are no longer reported resulting in limit recent comparisons. Hospitalisation crashes have steadily increased in recent years, possibly indicating more severe incidents or improved medical reporting. In contrast, medical treatment cases have declined, which may reflect better emergency care or revised classification methods.

To conduct this analysis, I created a dual-axis chart in Tableau. The main line represents total crashes per year, while coloured lines display severity categories, fatal, hospitalisation, medical treatment, minor injury, and property damage.

I used yearly data to highlight long-term trends without excess fluctuation. Each severity type was colour-coded, with darker tones for more serious outcomes to support easy comparison. I also added annotations to mark key events, such as the 2010 reporting change and the COVID-19 period. The chart includes labels, a legend, and interactive tooltips to enhance clarity and engagement.

4.2 Geographic Map

Geographic Hotspots of Road Crashes in Queensland



Figure 4.2 Geographic Hotspots of Road Crashes in Queensland (2001-2023)

Figure 4.2 highlights the geographic hotspots of road crashes across Queensland, using a heatmap by local police district. The visual clearly shows that crash incidents are heavily concentrated in the southeast, particularly in urban centres.

South Brisbane had the highest number of crashes (62,046 or 15.78%), followed by North Brisbane (55,076 or 14.01%) and the Gold Coast (42,667 or 10.85%). These three districts account for over 40% of all recorded crashes, reflecting the strong link between urban density and crash frequency.

This map offers valuable insights for transport authorities, identifying areas that may require targeted safety measures, infrastructure upgrades, or enhanced enforcement. The findings support region-specific strategies over one-size-fits-all policies.

To create this geographic analysis, I used a density heatmap, plotting crash locations using Crash Latitude and Crash Longitude. Colour intensity reflects crash volume, with warmer tones like red and orange highlighting high-density zones.

I applied a Mapbox base layer, adjusted the zoom to focus on Queensland, and fine-tuned the density setting for a balanced view. Labels were added to key districts (South Brisbane, North Brisbane, and the Gold Coast) to show total crash counts and percentages. This design supports both technical interpretation and easy understanding by general audiences.

4.3 Crash Severity by Speed Zone and Crash Type

Severity by Crash Type and Speed Limit

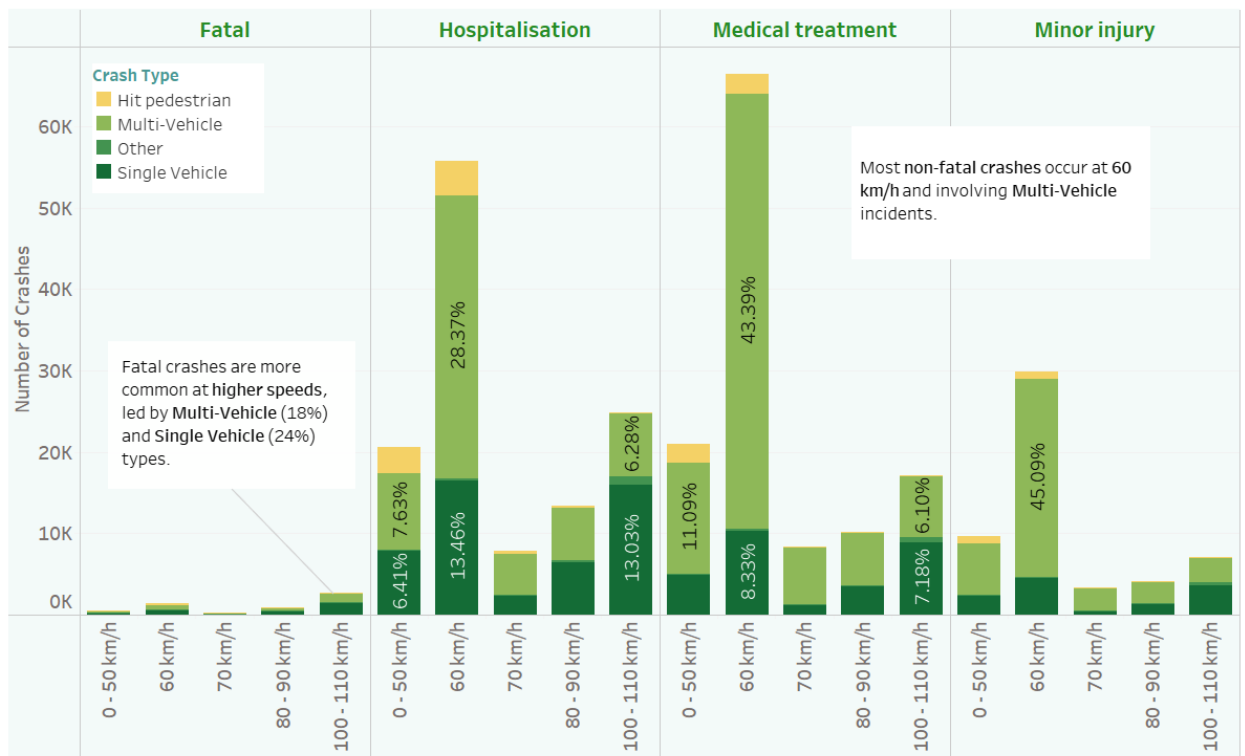


Figure 4.3 Severity by Crash Type and Speed Limit

Figure 4.3 illustrates the relationship between crash severity, crash type, and posted speed limits. The visual is divided by severity level, highlighting how crash patterns vary across speed environments.

Fatal crashes are more common at higher speeds, especially in 100-110 km/h zones, with significant shares involving Single Vehicle (24%) and Multi-Vehicle (18%) incidents. This suggests that high-speed loss of control contributes to most severe outcomes. In

contrast, most non-fatal crashes, including hospitalisation and medical treatment cases, occur in 60 km/h zones, typically in urban areas, and are dominated by Multi-Vehicle incidents (28-44%). Minor injuries also peak at 60 km/h, reinforcing the pattern that lower-speed environments produce more frequent but less severe crashes. These findings support targeted road safety measures by zone.

To create this visualisation, I built a stacked bar chart by placing Crash Severity on Columns and Crash Speed Limit on the x-axis. Crash Ref Number was added to Rows to count total crashes, and Crash Type was assigned to Colour to break down each bar by category, Single Vehicle, Multi-Vehicle, Hit Pedestrian, and Other. Also, I enabled percentage labels to show relative proportions and added annotations to highlight key insights. This format allowed me to compare severity levels and crash types across speed zones clearly, supporting targeted interpretation.

4.4 Crash Severity by Day of Week

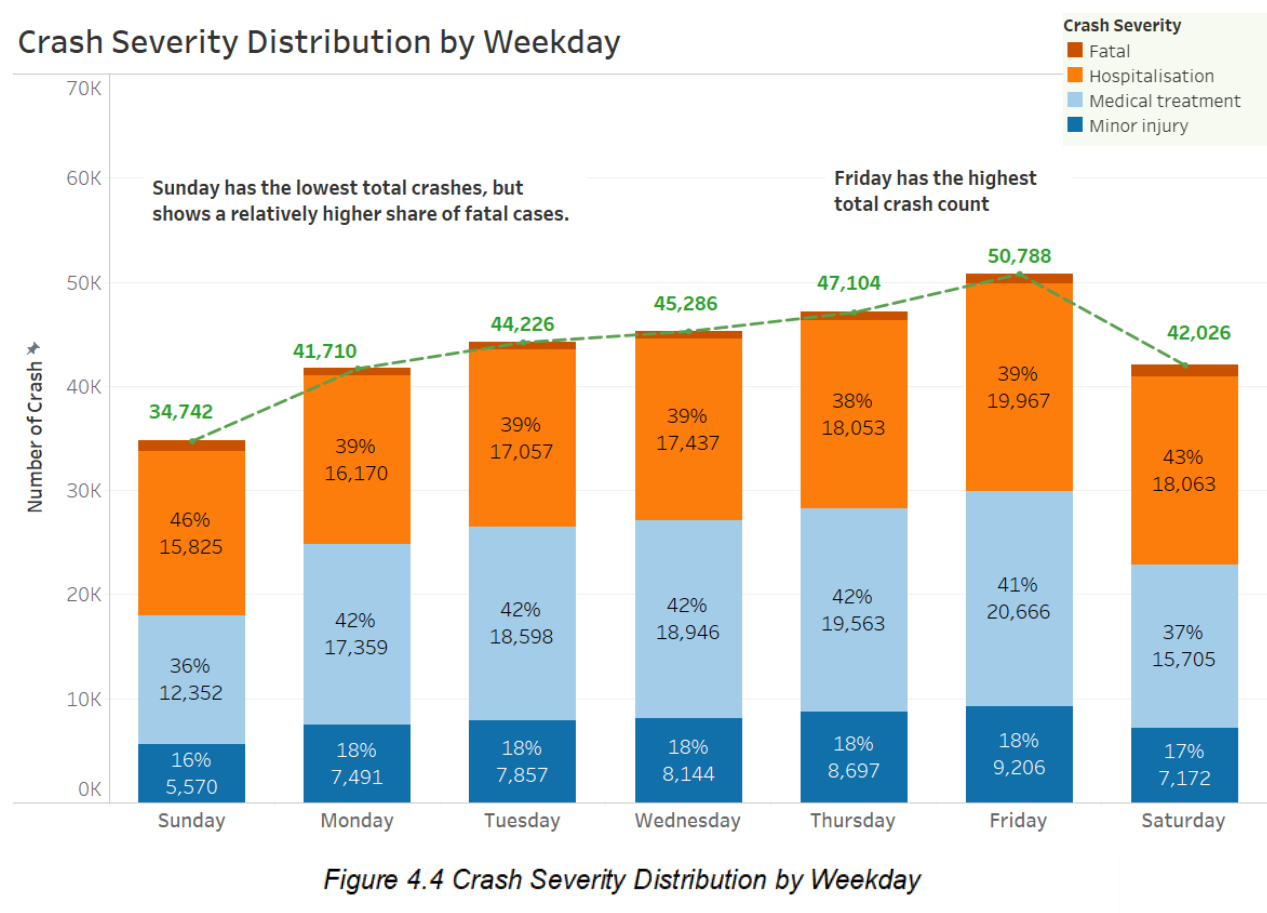


Figure 4.4 is a stacked bar and line combination chart showing the distribution of road crash severity by weekday, with total crash counts displayed as a green dashed line.

Friday records the highest total crashes (50,788), mainly involving medical treatment (41%) and hospitalisation (39%). Crash counts steadily increase from Sunday to Friday, reflecting typical weekday traffic volume. In contrast, Sunday has the fewest crashes (34,742) but a higher proportion of hospitalisation (46%) and fatal cases, suggesting weekend crashes may involve riskier behaviours like speeding. Severity proportions remain consistent during weekdays, with medical treatment and hospitalisation making up most incidents.

This dual format chart effectively highlights both the volume and severity composition of crashes, supporting data-driven decisions in road safety planning and enforcement. To create this visualization, I built a stacked bar and line chart by placed Crash Day of Week on the x-axis and added Crash Ref Number to Rows to count total crashes. Crash Severity field was assigned to the Colour mark to stack the bars by severity type (Fatal, Hospitalisation, Medical Treatment, and Minor Injury). A dual-axis line chart was created to show total crash counts across weekdays. I enabled percentage labels within each bar and included annotations to highlight key observations. This format helps compare both crash volume and severity patterns throughout the week.

4.5 Crash Severity by Lighting and Weather Conditions

Crash Severity by Combined Lighting and Atmospheric Conditions

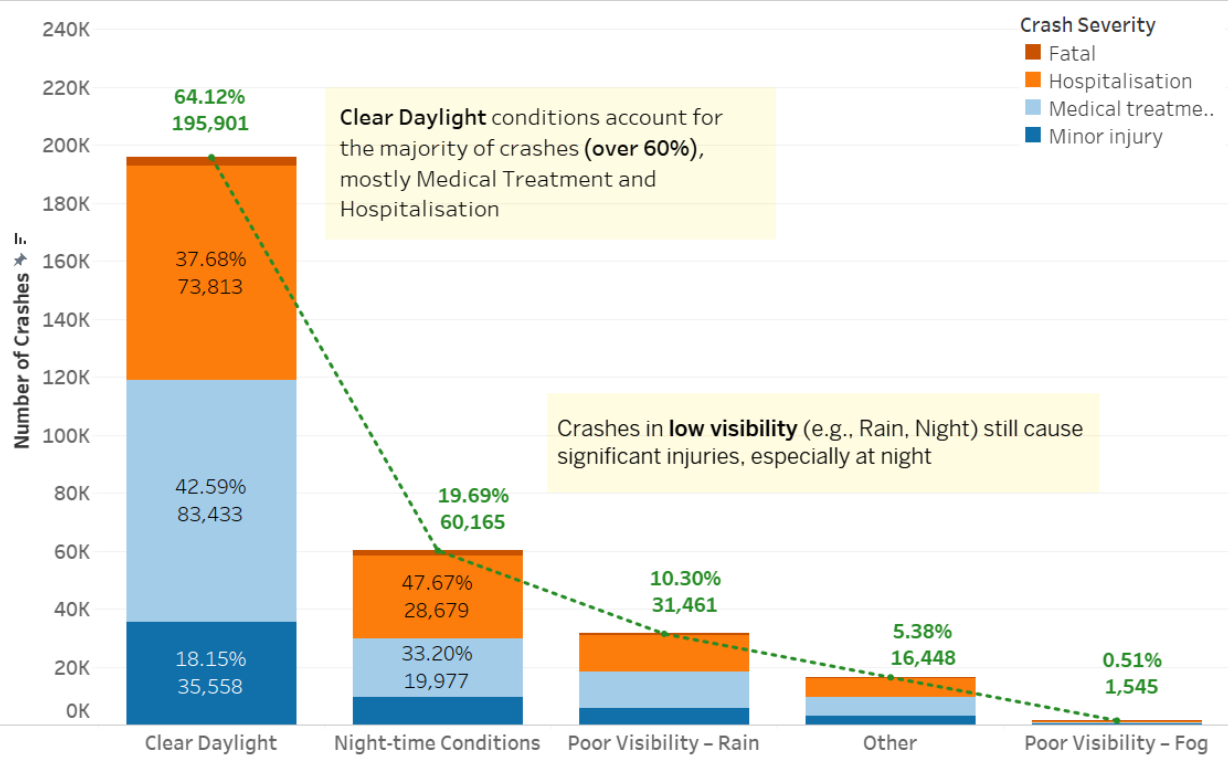


Figure 4.5 Crash Severity by Combined Lighting and Atmosphere Conditions

Figure 4.5 is a stacked bar and line combination chart displaying crash severity by combined lighting and atmospheric conditions, with total crash counts shown as a green dashed line.

Clear daylight accounts for the highest number of crashes (195,901 or 64.12%), where most incidents involve medical treatment (42.59%) and hospitalisation (37.68%). This reflects high exposure during daytime driving. Under night-time conditions, hospitalisation cases make up 47.67%, the largest share within that group, highlighting the severity of low-visibility environments. Similarly, rain-related crashes see over 40% of cases resulting in either hospitalisation or medical treatment.

Although fog-related crashes are rare (0.51%), they still span across all severity types, including fatal cases. This pattern suggests poor lighting and weather increase injury risk even if crash volume is lower, underscoring the need for better visibility management and public safety awareness in these conditions.

To create this visualisation, I created a calculated field combining Crash Lighting and Atmospheric Conditions into five grouped environment types. This field was placed on the x-axis, with Crash Ref Number on Rows to count total incidents. Crash Severity was applied to Colour to segment each bar by severity level. A stacked bar chart was used to show severity distribution, and a dual-axis line was added to represent total crash counts. Percentage labels and annotations were included to highlight key differences.

5. Ethical Considerations in Data Visualisation

- **Transparency:** Clearly labelled axes, defined groupings (e.g., *Combined_Environment_Group*), and key annotations were used to prevent misinterpretation. Misleading visual scales were avoided to ensure honest representation which crucial in industry for informed decision-making.
- **Accuracy:** Ensuring through data cleaning, such as removing inconsistent time-lighting entries and null geographic points, to enhance reliability where clean and consistent data is essential for generating actionable insights in professional contexts.
- **Privacy and Responsible Use of Data:** Although the dataset contained no personally identifiable information, I avoided overemphasising specific crash hotspots. Broader regional patterns were shown to prevent community stigmatisation and align with industry expectations under privacy standards such as GDPR.

6. Conclusion

This analysis of Queensland road crash data revealed key patterns across time, location, severity, and environmental conditions. Urban centres such as Brisbane and the Gold Coast recorded the highest crash volumes, especially in 60 km/h zones involving multi-vehicle crashes likely due to stop-and-go traffic and driver impatience. Crashes were most frequent during daylight hours, aligning with commuter peaks for workers and students. Fatal crashes, though less common, were concentrated in high-speed zones (100-110 km/h), while poor lighting and adverse weather were linked to higher injury severity. Time-based trends showed a decline in crashes after the 2010 reporting change and during COVID-19. These insights support targeted safety strategies, including speed control, infrastructure upgrades, and public education tailored to urban environments and high-risk conditions.